TECHNICAL REPORT

# French→English Transformer from scratch: where it generalizes and where it breaks

Encoder-decoder trained from scratch on filtered opus-100; evaluated in-domain, on long sentences, and on literature it never saw.

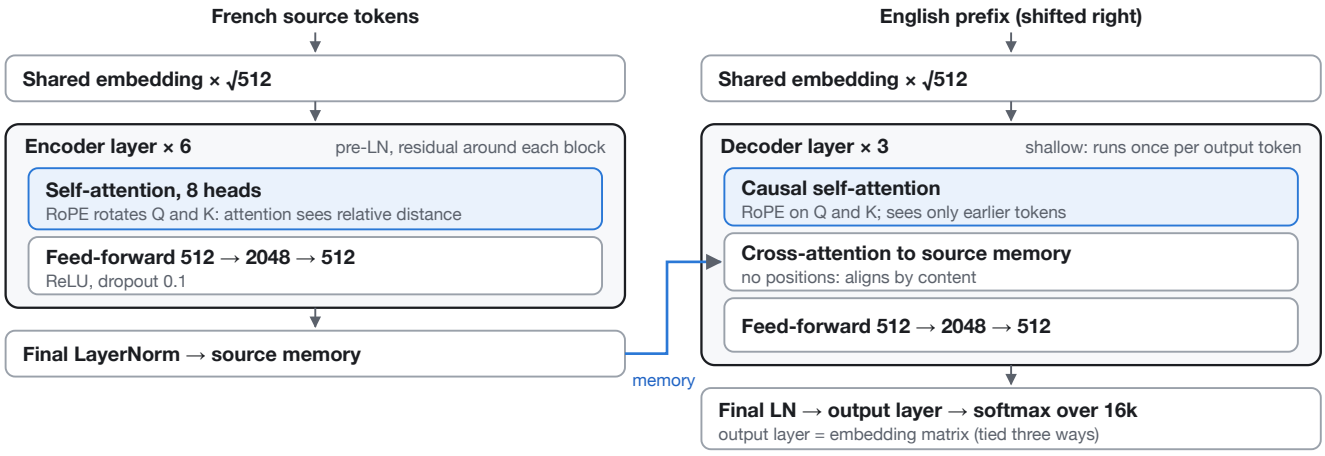
GitHub bad2212/transfromer-model | Hugging Face Badalt/fr-en-transformer-scratch | W&B run 5ks4crpz | W&B report dashboards

| 39.94 | 29.8 / 46.4 | 34.7 / 60.0 | 21.9 / 43.4 | 39.7M |
|---|---|---|---|---|
| Dev OVERALL score | Seen · BLEU / chrF | Long · BLEU / chrF | Books (unseen) · BLEU / chrF | Params · 3.6 h, one T4 |

**KEY FINDINGS**

1. **Length is not the failure mode; domain is.** Literature trails in-domain at every sentence length.
2. **The dev gap needs more data to be trusted.** On 60 sentences its 95% CI includes zero; on 1,000 per domain it is 31.6 vs 15.8 BLEU.
3. **Register explains more than vocabulary.** Books sentences with no rare word still trail by 14.1 BLEU; rare words cost more on top.

## 1 Architecture decisions




39.7M parameters. Positions enter only through RoPE inside the two self-attentions; the encoder's output (memory) feeds the cross-attention of every decoder layer; one 16k × 512 matrix serves as both embeddings and the output layer.

| Choice | Why | Rejected alternative |
|---|---|---|
| **Encoder-decoder [1]** | Reads the source bidirectionally; cross-attention aligns target to source words | Decoder-only: causal view of the source, sequences 2× longer |
| **6 enc / 3 dec, d 512 [2]** | **Measured** on the T4: 6/6 ran at 11.5k target tok/s, 6/3 at 13.6k, so ~20% more updates in budget | 6/6 base: slightly better per step, 15% slower |
| **Pre-LayerNorm [3]** | Stable gradients at a high peak LR (7e-4) | Post-LN: diverges easily early |
| **RoPE, self-attention only [4]** | Relative positions; cross-attention aligns by content. Not chosen for extrapolation: *long* is within the trained range | Sinusoidal (ablation config), ALiBi [5] |
| **Tied embeddings [6]** | One matrix for encoder input, decoder input and output layer; saves 16.4M params | Separate vocabularies |
| **SentencePiece 16k [7, 8]** | Joint unigram vocab, byte fallback (no <unk>), source-side subword sampling α=0.1 | BPE [9], 32k vocab, characters |
| **Beam 5, α=1.8 [10]** | Tuned on dev; chrF (β=2) and BLEU's brevity penalty both reward full-length output | Greedy: −1.4 OVERALL |
| **Own PyTorch code** | RoPE inside attention, pre-LN, tying; ~170 lines, unit-tested | `nn.Transformer`: no RoPE hook |

## 2 Data and training

**Data.** opus-100 en-fr *train* only, no subsampling. Unicode and quote normalisation (books sources use ’, their references use '), then filtering kept **923,274 of 1,000,000** pairs: 32,548 duplicates, 21,617 bad length ratios, 14,770 untranslated copies, 6,602 empty, 1,186 over-long, 3 dev/test overlaps; **922,994** were used in training (280 exceed the 256-subword cap). opus_books was never used for training or tuning. Dev slices come from opus-100 validation (*seen*), opus-100 test (*long*) and opus_books (*unseen*); the logged validation loss includes the 60 *seen* sentences but drove no choice.

**Training.** Label smoothing 0.1, AdamW (0.9, 0.98), 4k warmup + inverse-sqrt, ~24k tokens per update, fp16, seed 42. Stopped at step 9,000 (~8.3 epochs, 3.6 of a planned 5.5 h) when the free-Colab GPU quota ran out, so the model is **under-trained**. Final weights average checkpoints 7k–9k [11] (greedy 38.49 vs 38.14 for 9k alone).

| Step | Val loss | Dev OVERALL (greedy) |
|---|---|---|
| 2,000 | 4.11 | 14.4 |
| 4,000 | 2.65 | 30.8 |
| 6,000 | 2.28 | 35.4 |
| 8,000 | 2.14 | 37.6 |

| Decoding (dev) | OVERALL | BLEU | chrF |
|---|---|---|---|
| Greedy | 38.49 | 28.45 | 46.81 |
| Beam 4, α 1.4 | 39.47 | 30.00 | 47.50 |
| Beam 5, α 1.4 | 39.83 | 30.23 | 47.77 |
| Beam 5, α 1.8 — chosen | 39.94 | 30.23 | 47.91 |
| Beam 5, α 2.2 | 39.83 | 30.09 | 47.85 |

Full 11-point grid in the appendix. Gains above α 1.4 are within noise.

## 3 Dev results by slice

| Slice | n | BLEU | 95% CI | chrF | 95% CI |
|---|---|---|---|---|---|
| **seen** | 60 | **29.8** | 21.9 to 38.0 | **46.4** | 40.4 to 52.4 |
| **long** | 30 | **34.7** | 24.9 to 43.9 | **60.0** | 53.9 to 66.1 |
| **unseen domain** | 60 | **21.9** | 17.1 to 26.2 | **43.4** | 39.3 to 48.0 |
| **All** | **150** | **30.2** | | **47.9** | |

OVERALL = 0.4·BLEU + 0.4·chrF + 0.2·$\text{chrF}_{\text{unseen}}$ = **39.94**. Bootstrap CIs use 1,000 resamples. The seen–unseen gap is 7.9 BLEU (CI −1.8 to 17.5) and 3.0 chrF (CI −4.5 to 10.6): **not significant on 60 sentences**, so section 4 re-measures it on 1,000. *Long* beats *seen* because long opus-100 sentences are formulaic official text, while short ones are colloquial subtitles with loose references (about 4 of 150 dev pairs are misaligned).

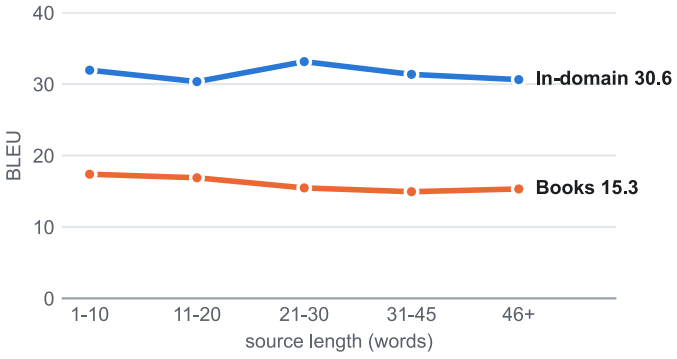
## 4 Generalization: evidence and hypothesis

| Sample (n=1,000 each, greedy) | BLEU | chrF | Unseen words | Rare words | Ref. loss | Out/ref length |
|---|---|---|---|---|---|---|
| ■ **opus-100 test (in-domain)** | 31.6 | 52.0 | 1.1% | 2.2% | 2.05 | 0.94 |
| ■ **opus_books sample (unseen)** | 15.8 | 37.9 | 1.9% | 5.1% | 2.88 | 0.90 |

In-domain (opus-100 test) Books (opus_books)

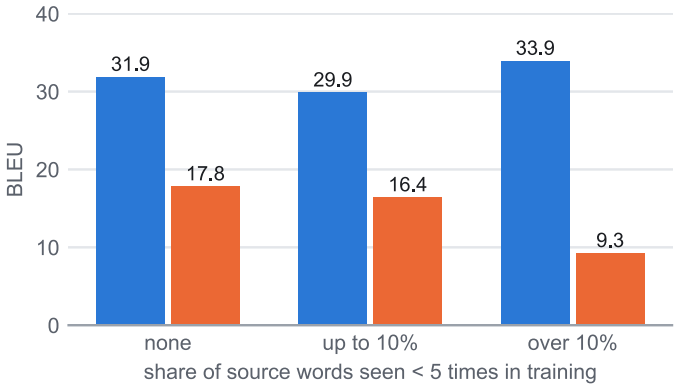
**Books trail at every length**

BLEU by source length; in-domain stays flat even past 45 words




**Rare words hurt only in books**

BLEU by share of rare source words; rare-free books already lag




1. **Register is the largest share.** Books sentences with no rare word score 17.8 vs 31.9 BLEU in-domain, and the model's loss on the correct reference is 2.88 vs 2.05 nats/token (perplexity 17.9 vs 7.8): trained on subtitles and official text, it finds 19th-century narrative English improbable.
2. **Rare vocabulary adds to it.** Books have 1.8× the never-seen-word rate and 2.4× the rare-word rate; the 190 books sentences with over 10% rare words fall to 9.3 BLEU.
3. **Freer references and shorter outputs.** Output/reference length is 0.90 on books vs 0.94, so BLEU's brevity penalty bites harder. Typography is ruled out by normalisation.

## 5 Where it fails

| Failure | Examples and frequency |
|---|---|
| **Names, rare words** | Subword pieces recombine into a frequent word: *Le grand Meaulnes* → “The Great Meals”, *roulotte* → “roller”, *grognait* → “grew up” |
| **Literary word senses** | The common sense wins: *cils* (eyelashes) → “cloaks”, *fers* (forceps) → “irons”. Past tenses mostly right (*répondit* → “answered”), with slips (*jeta* → “is throwing”) |
| **Repetition loops** | Rare: 0.5% of in-domain and 0.4% of books outputs repeat a 3-gram three or more times |
| **Very long inputs** | Inputs over 128 subword tokens are split at sentence ends: on 23 such sentences, chrF 41.2 split vs 39.5 unsplit |
| **Reference noise** | Misaligned opus-100 pairs and books references with stray quotes or chapter headings cap the scores |

## 6 Challenges and how they were resolved

| Problem | Resolution |
|---|---|
| **Colab disconnects; GPU quota ended training at 3.6 h** | Atomic full-state checkpoints, deterministic batch order, wall-clock budget across sessions, same W&B run. A planned kill-and-resume test caught a bug (RNG state moved to GPU) before the real run |
| **T4 slower than planned** | Switched 6/6 → 6/3 on a throughput threshold fixed before measuring |
| **Noisy data; curly vs straight quotes** | Filtering (7.7% removed); one normaliser on inputs and outputs (one ’ costs ~20 chrF on a sentence) |
| **Decoding stalls on one long input** | Length-sorted batches capped at 2,048 source tokens; outputs unchanged |
| **Tokenizer differs across machines** | 12 of 16k pieces changed with the thread count; pinned to 2 threads, and the trained tokenizer ships with the model |
| **Beam-search and splitter edge cases** | Unit tests: greedy = beam(1), overfit-and-recover, abbreviation-aware splitting |

## 7 How the results were validated

| Check | Evidence |
|---|---|
| **Model correctness** | 5 unit tests: causal mask, padding invariance (RoPE and sinusoidal), a tiny model memorises 8 pairs and greedy, beam(1) and beam(4) all recover them, normalisation, sentence splitting |
| **Metric fidelity** | Scoring imports the official `score.py`; identical OVERALL (39.94) to running it directly |
| **No leakage** | Dev/test sources removed from training; opus_books used only after every choice was fixed; all tuning on dev |
| **Reproducibility** | Identical filter counts on two machines; seeded batch order; one command reproduces the run |
| **Uncertainty** | Bootstrap CIs on every dev slice; the gap re-measured on 1,000 sentences per domain |
| **Claims vs data** | Each failure claim counted on 2,000 outputs before writing; two over-strong draft claims were corrected |
| **Deliverables** | Submission matches all 330 ids; Hugging Face weights match by SHA-256 and load with 3 dependencies. All 21 checks are in the appendix |

## 8 Next steps with more time or compute

- **Train to convergence** (dev still rising at 9k steps) with a cosine tail, and several seeds for variance.
- **Close the domain gap:** back-translate monolingual English literature (not opus_books) [12], domain tags, a copy mechanism for names.
- **Run the prepared ablations** (RoPE vs sinusoidal, 16k vs 32k vocab) and stronger data filtering [13].
- **Decoding:** KV cache plus MBR with chrF as the utility; report sacreBLEU alongside the challenge scorer.

## References


[1] Vaswani et al. 2017, Attention Is All You Need
[2] Kasai et al. 2021, Deep Encoder, Shallow Decoder
[3] Xiong et al. 2020, On Layer Normalization in the Transformer
[4] Su et al. 2021, RoFormer: Rotary Position Embedding
[5] Press et al. 2022, Train Short, Test Long (ALiBi)
[6] Press & Wolf 2017, Using the Output Embedding
[7] Kudo & Richardson 2018, SentencePiece
[8] Kudo 2018, Subword Regularization
[9] Sennrich et al. 2016, NMT of Rare Words with Subword Units
[10] Wu et al. 2016, Google's NMT System
[11] Popel & Bojar 2018, Training Tips for the Transformer
[12] Sennrich et al. 2016, NMT with Monolingual Data
[13] Junczys-Dowmunt 2018, Dual Conditional Cross-Entropy Filtering